

"I'm writing this for those novices out there who are itching to develop Android apps, but don't yet know how to. Until a few weeks ago, I too had no idea about creating Android apps, but I really wanted to develop my own mobile apps. I did some reading on the Internet, and found out that Android apps are developed using Java. I was a little disheartened at first because I had never learnt Java or, for that matter, any Object Oriented Programming language. Nonetheless, I found the time in my summer vacation to attend a few classes on Java at a local teaching centre. A few weeks into it, I discovered that Java is extremely intuitive and easy to grasp.

"Earlier experience in Java would definitely be a plus point, but don't be disheartened if you don't know any Java at all. There are plenty of resources online to help you learn, and using a book wouldn't be a bad idea as well. I must stress that getting your syntax bang on is not as important as getting the hang of the basics and the fundamental concepts. You should have a solid understanding of OO concepts such as object creation, inheritance, overriding, interfaces, collections, packages, etc. These might sound like Greek and Latin, but once you get started on Java, this will be your bread and butter.

"A lot of people think that knowing XML is crucial in building Android apps, but the truth is that you can build whatever you want with no previous knowledge of XML. This is basically because the XML elements are integrated as a drag and drop feature in the Android environment. Also, XML is very simple and you will definitely pick it up while building an app. Learning Android is merely about getting to know a few additional concepts. First, visit *developer.android.com*. You will find instructions there to install and set up your SDK (what you'll need to develop the apps). This contains half-built Android programs, in the sense that it contains incomplete classes that you have merely to extend (again, don't worry about these terms, you'll understand them soon enough) to build your apps. This way, it becomes very simple as you don't have to start from scratch.

"There are plenty of resources online to help you learn Android. Udacity (www.udacity.org) and Coursera (www.coursera.org) offer courses on mobile app development. The course in Udacity, named 'Developing Android apps', is interactive and is centred on the purpose of making you build your own weather app. The entire course involves the steps to build that app, and you are expected to do so on your own. The idea is that, from this, you will develop your skillsets to develop your own Android apps. This is a very nice concept, and provides a hands-on learning experience. There is also another course available from Coursera titled, 'Creative, Serious and Playful Science of Android Apps'. This requires no previous knowledge of Java. You may find this useful as well.

"The two IDEs that you can use for Android development are Android Studio and Eclipse. While both are very similar in many ways, I use Eclipse, though this

is entirely your choice. As Android Studio is still in the early stages of development, and as only the beta version is available for use, I would recommend using Eclipse (www.eclipse.org). Android apps can be made on Eclipse by downloading the necessary plugins. The complete package that you need to install, including Eclipse with the necessary plugins, as well as the SDK manager (an app that manages the downloading and installation of build tools and libraries) is available on *developer.android.com*, and is called the ADT (Android Developer Tools) bundle (JDK must be pre-installed). The IDE contains Java classes and XML files, that are made to interact with the help of Android manifest, which is an XML file.

"Essentially, each screen of an Android project is called an activity, and every activity has a backing XML file and Java class. Android apps follow the MVC (model view controller) design pattern. The XML file, which is for UI, acts as the view, whereas the backing Java classes are for the model and controller. If you do not wish to do the XML part, the UI can be designed by the intuitive dragging and dropping of UI elements (called views), and the IDE will automatically make changes in the XML file.

"Debugging is made simple with the help of LogCat, a debugging tool that shows what exactly is going on at every step in the runtime of the application. The app that you create can be run either using an emulator or on your Android device. I would recommend the latter, as the emulator is incredibly slow in loading. If you want to run on your Android device, make sure that you install the device driver. Also, if your Android version is 4.2 or higher, you will need to turn on developer options. You can do this by going to 'About phone' in 'Settings' and then clicking on 'Build version'. However, if you do not have an Android device, you will have to create an emulator on AVD (Android Virtual Device) manager. Simple apps such as a calculator and apps to print out your name are surprisingly easy to build. Now go, get started on your Android adventure!"

Thank you, Ankith, for sharing your experience on learning Android app development!

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming! 

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